AI RESUME ANALYZER AND JOB RECOMMENDATION SYSTEM

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Tech Used: Python, Streamlit

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1. PROJECT OVERVIEW

The AI Resume Analyzer and Job Recommendation System is a simple web application that reads a resume and checks which technical skills are mentioned in it. It then compares those skills with a list of job roles and suggests the roles that best match the resume.

The application can also show skills that may be missing for a selected role and provide a basic learning plan to help the user improve their skills.

The project is built using Streamlit, which allows users to upload their resumes and view the results on a single page. Users can also download a report in CSV format.

2. PROBLEM STATEMENT AND OBJECTIVES

Students may sometimes find it difficult to understand which jobs match the skills listed in their resumes. This project aims to make that comparison easier and help users explore possible career options.

The application is intended only as a guide for career exploration and should not be used to make actual hiring decisions.

2.1 Objectives

The main objectives of the project are:

- To accept resumes in PDF and DOCX formats.
- To extract text from resumes and clean up spacing and letter case.
- To identify technical skills using a prepared list of keywords.
- To compare resumes with ten sample job roles.
- To display the top three matching job roles.
- To identify skills that were not found for a selected role.
- To suggest learning topics for improving missing skills.
- To allow users to download the results as a CSV report.

3. TOOLS AND TECHNOLOGIES USED

The following tools and libraries are used in the project.

Tool or Library	Purpose
Python	Main programming language used to develop the project.
Streamlit	Creates the web interface for uploading resumes and displaying results.
pypdf	Extracts text from PDF files.
python-docx	Reads text from DOCX files.
pandas	Handles CSV files and tabular data.
NumPy	Supports numerical operations.
Regular Expressions	Helps clean and process resume text.
scikit-learn	Uses TF-IDF and cosine similarity to compare text.
Plotly	Displays a chart of job matching scores.

4. APPLICATION WORKFLOW

The application follows these steps:

Sno	Description
1	The user uploads a resume in PDF or DOCX format.
2	The application extracts the text and cleans its spacing and letter case.
3	It searches for skills using a prepared list of keywords.
4	It compares the resume text with descriptions of different job roles.
5	It ranks the job roles and displays the top three matches.
6	The user selects a role to view missing skills, learning suggestions, and a downloadable report.

5. MAIN PROGRAM FILES

The project is divided into different Python files, with each file handling a specific task.

File Name	Purpose
app.py	Runs the Streamlit interface and handles report downloads.
resume_parser.py	Extracts text from PDF pages and DOCX paragraphs.
text_cleaner.py	Cleans resume text while keeping useful symbols such as + and #.
skill_extractor.py	Finds skills from the prepared skill list and groups them by category.
job_matcher.py	Compares resumes with job roles and checks their skill requirements.
roadmap_generator.py	Suggests learning topics for up to four skills that were not found.

6. DATA USED IN THE PROJECT

The project uses two CSV files containing the information needed for skill detection and job matching.

6.1 skill_dictionary.csv

This file contains around 40 technical skills along with their categories. The application uses this list to identify skills mentioned in a resume.

6.2 job_roles.csv

This file contains ten sample job roles and the skills associated with each role.

The job roles included are:

1. Data Analyst
2. Machine Learning Engineer
3. AI Engineer
4. NLP Engineer
5. Computer Vision Engineer
6. Web Developer
7. Backend Developer
8. Data Scientist
9. Cloud Engineer
10. Full Stack Developer

These roles and their skill requirements are examples created for the project. They are not a complete or verified list of the skills required by employers.

7. HOW JOB MATCHING WORKS

The application uses TF-IDF and cosine similarity to compare the resume with different job roles.

TF-IDF converts text into numerical values based on the importance of words in the text. Cosine similarity then measures how similar the resume text is to the descriptions of the available job roles.

After the comparison, the application calculates matching scores and sorts the roles from highest to lowest. The top three results are displayed to the user.

The matching score only indicates how similar the text is. It does not represent the exact percentage of skills a person possesses, nor does it predict whether they will get hired. The results can also be affected by the wording used in the resume and how the text is processed.

8. SKILL GAPS AND LEARNING PLAN

After selecting a job role, the application compares the skills detected in the resume with the skills listed for that role.

It then displays the keywords that were not found in the resume. Based on these missing keywords, the application suggests learning topics, with one topic per week for up to four missing skills.

This feature helps users understand which areas they could focus on while preparing for a particular career.

However, a skill not being detected does not necessarily mean that the user does not know it. The skill may have been written differently in the resume, or the application may not recognise the wording.

9. TESTING AND CURRENT STATUS

The application was reported to be working after the column names in the CSV files were corrected. However, a completed testing sheet and measured test results were not available when this report was prepared.

The following tests still need to be carried out and recorded.

Test	What Needs to Be Checked
PDF and DOCX files	Check whether text can be extracted from both file formats.
Job role suggestions	Test sample resumes for data analysis, machine learning, and web development roles and record the results.
Missing skills	Add a skill to a sample resume and check whether it disappears from the missing-skill list.
Unusual files	Test image-only PDFs, resumes with no recognised skills, and unsupported file types.
Skill detection	Check for incorrect matches, such as Java being detected inside JavaScript.
CSV download	Open the downloaded file and check its roles, scores, and skill lists.

These tests are important for identifying errors and improving the reliability of the application.

10. HOW TO RUN THE PROJECT

To run the application, Python must be installed on the system along with the required project files.

Step 1: Install the Required Packages

Open a terminal in the project folder and run the following command:

```
python -m pip install -r requirements.txt
```

This installs the libraries listed in the requirements.txt file.

Step 2: Start the Application

After installing the packages, run:

```
python -m streamlit run app.py
```

Step 3: Open the Application

A local address should appear in the terminal. Open that address in a web browser to access the application.

The user can then upload a resume containing selectable text and view the analysis results.

11. LIMITATIONS OF THE PROJECT

The project has a few limitations that could be addressed in future versions.

- **Keyword-based detection:** The application searches for specific keywords, so it may sometimes identify skills incorrectly.
- **Limited skill recognition:** It may not recognise every possible way of writing a skill.
- **Image-only PDFs:** The current version cannot extract text from PDFs that contain only images.
- **DOCX tables:** Information stored in tables inside DOCX files may be missed.
- **Sample job requirements:** The job roles use prepared skill lists rather than verified requirements from employers.
- **No live job vacancies:** The application does not search the internet for current job openings.
- **Limited resume evaluation:** Education, work experience, and project quality are not evaluated separately.
- **Testing requirements:** More complete testing and file-size checks would help improve the application.

12. PRIVACY AND RESPONSIBLE USE

For demonstrations, it is safer to use fictional resumes or remove personal information before uploading a resume.

The application should not be used to automatically accept or reject job applicants. Personal details such as age, gender, religion, nationality, disability, and marital status should not be used to judge a candidate's suitability for a job.

The current code does not explicitly save uploaded resumes to disk or send them to an external AI service. However, if the application is published online, a clear privacy notice, suitable upload limits, and careful handling of logs and downloaded reports would be necessary.

Real resumes should not be uploaded to a public code repository because they may contain personal information.

13. FUTURE IMPROVEMENTS

If I continue working on this project, there are several improvements I would like to make.

- Improve skill detection so that it recognises complete skill names more accurately.
- Add more alternative names and keywords for technical skills.
- Prevent similar words from being confused during skill detection.
- Improve text extraction from tables in DOCX files.
- Display a separate measure of how many required skills were detected.
- Analyse different sections of a resume separately.
- Test the application more thoroughly and record the results.

These improvements could make the application more accurate and easier to use.

14. WHAT I LEARNED

While working on this project, I got to work with Python, file handling, text cleaning, CSV data, and Streamlit. I also learned how TF-IDF and cosine similarity can be used to compare text and suggest suitable job roles.

The project helped me understand how different Python libraries can work together to create a simple web application.

I also learned that matching scores need to be explained properly and that even if a program runs successfully, it still needs to be tested before its results can be trusted.

15. CONCLUSION

The AI Resume Analyzer and Job Recommendation System is a basic Python project that compares the skills mentioned in a resume with a prepared list of job roles.

It helps users explore which roles may match their current resume and identify skills they could learn in the future. The use of Streamlit makes the application simple to access, while TF-IDF and cosine similarity provide a basic method for comparing resume text with job descriptions.

Although my project has limitations, it provides a useful starting point for learning Python programming, text processing, file handling, and simple recommendation systems.

16. DOCUMENTATION AND REFERENCES

The following official documentation was used as a reference for the tools and libraries used in the project.

1. **pypdf Documentation:** https://pypdf.readthedocs.io/
2. **python-docx Documentation:** https://python-docx.readthedocs.io/
3. **scikit-learn Documentation:** https://scikit-learn.org/
4. **Streamlit Documentation:** https://docs.streamlit.io/